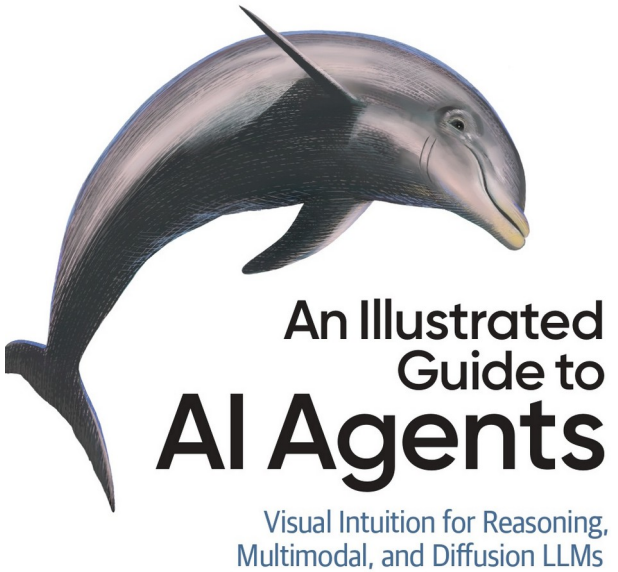
M5

Planning & ReAct

Module 5 — Making the agent autonomous



307307 — BI Methods for Generative AI

Agentic AI · University of Petra

Lecturer: _____ Date: _____

Learning Objectives

1. Explain task decomposition — breaking a goal into smaller steps.
2. Describe the ReAct loop: THOUGHT, ACTION, OBSERVATION, repeated until done.
3. Explain what makes an agent autonomous (a loop + a stopping rule).
4. Describe reflection (self-critique) and why it helps.
5. Recognise what lies beyond a single agent: evaluation, multi-agent, multimodal, coding.

Where We Are — the Final Piece

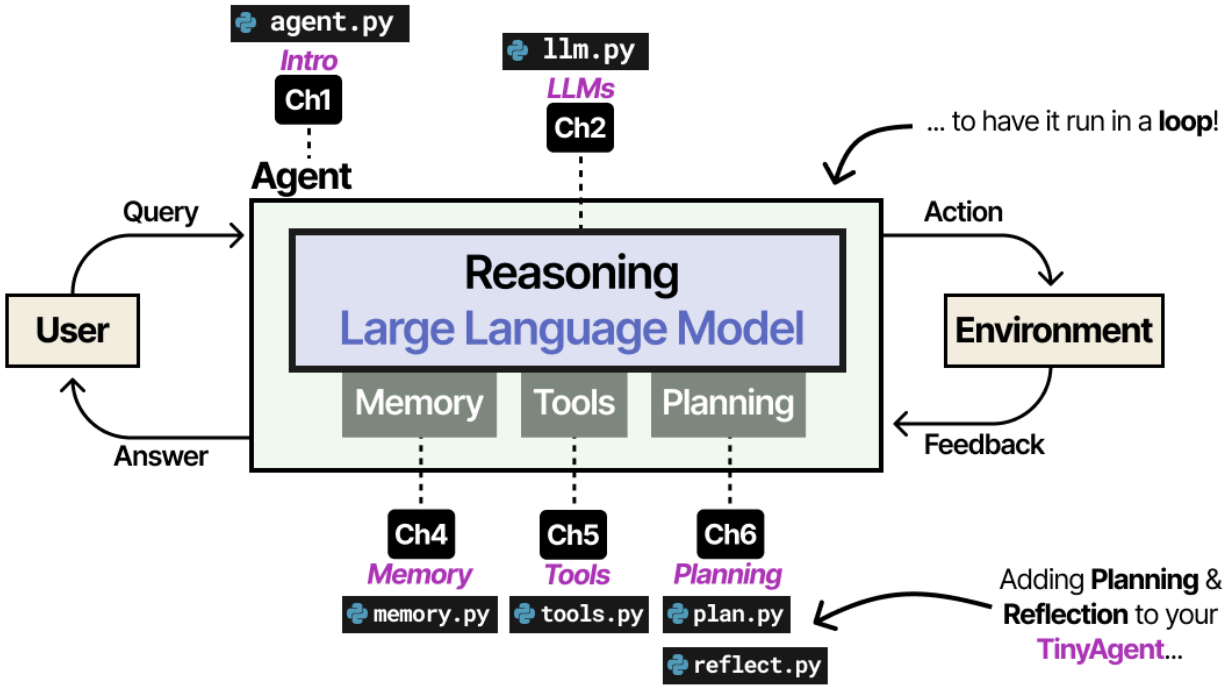


Figure from An Illustrated Guide to AI Agents (O'Reilly, 2026)

Think · Remember · Act

M2 - M4

Plan & loop (today)

M5 — autonomy

= a complete agent

the foundation is done

Planning — Decompose the Task

Real tasks take many steps.

Task decomposition means breaking a big goal into smaller, ordered sub-tasks, so the agent knows what to do next — and can revise the plan as it learns more.

A reasoning LLM does this naturally ("think step by step").

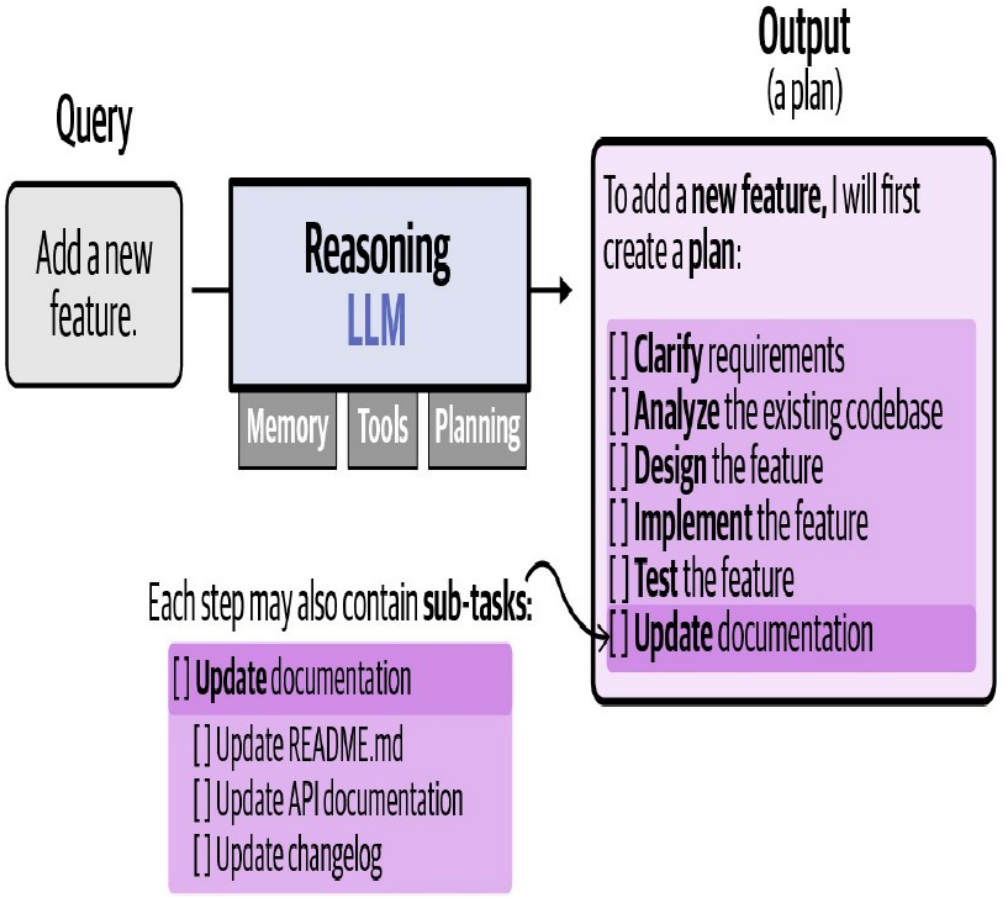


Figure 6-2, An Illustrated Guide to AI Agents (O'Reilly, 2026)

ReAct — Reason + Act in a Loop

The agent repeats a simple cycle until the task is done:

THOUGHT	reason about the next step
ACTION	call a tool (from Module 4)
OBSERVATION	we run it and feed the result back

...repeat, using each observation to decide the next thought, until a final answer.

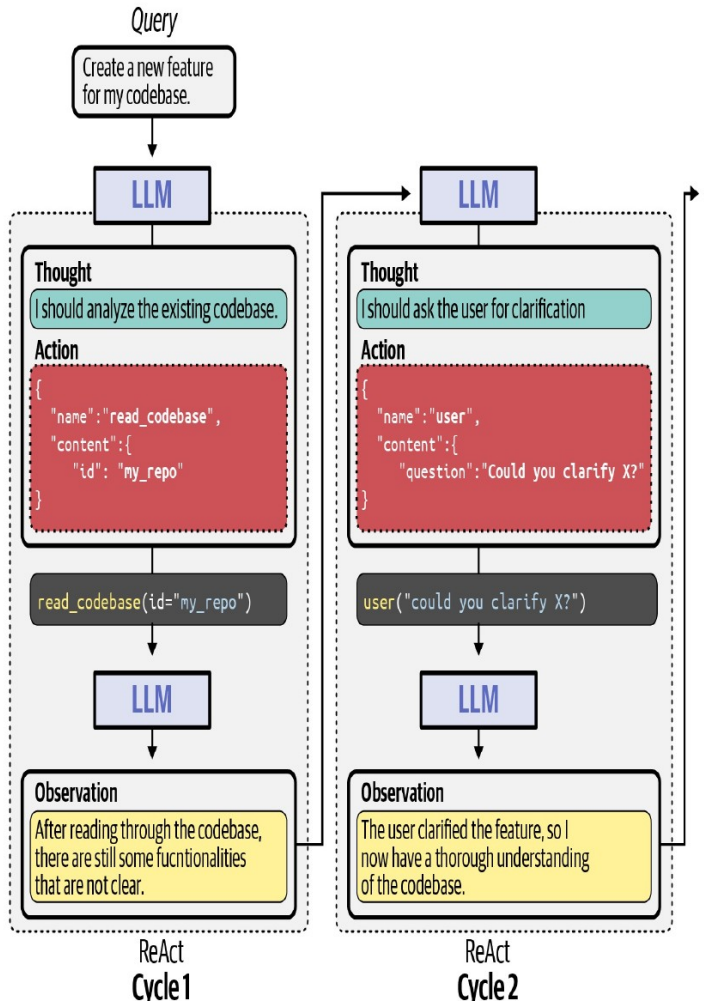


Figure 6-13 (O'Reilly, 2026)

What Makes It Autonomous

The “magic” is surprisingly simple — a loop. The agent keeps running THOUGHT -> ACTION -> OBSERVATION until it decides to stop.

A stopping rule

The agent signals it is done by calling a special final_answer.

max_steps

A cap on how many loops it may run — so a confused agent can never loop forever.

The complete single agent

Reasoning LLM (brain) + **Memory** (remembers its steps) + **Tools** (acts) + **Planning / ReAct** (sequences actions). Everything since Module 1 led here.

Reflection — Learning From Mistakes

Agents can get stuck or go wrong. Reflection has the agent critique its own work and course-correct.

Self-Refine

The LLM drafts, gives itself feedback, and refines — like a writer editing their own draft.

Reflexion

A richer version with separate roles: one acts, one evaluates, one writes feedback for the next try.

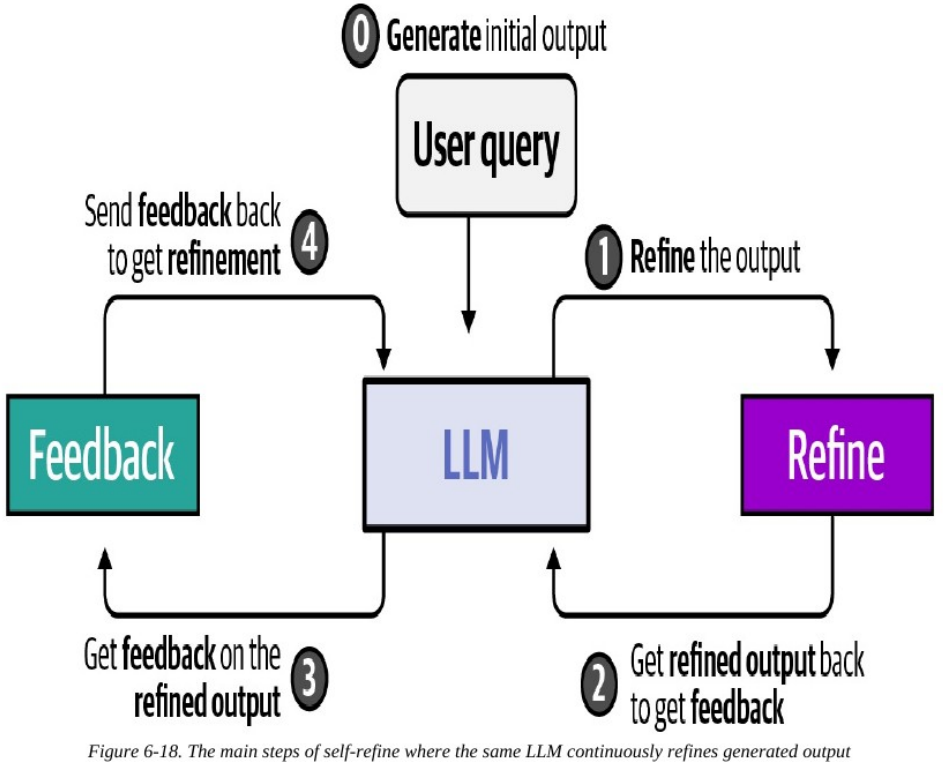


Figure 6-18. The main steps of self-refine where the same LLM continuously refines generated output

Figure 6-18 (self-refine), An Illustrated Guide to AI Agents (O'Reilly, 2026)

What You Will Build Today

1 The ReAct loop

a for-loop of THOUGHT -> ACTION -> OBSERVATION.

2 Stopping & cap

final_answer to stop; max_steps so it can't loop forever.

3 Multi-step task

(4.6 + 6.685) x 4, then subtract 3.14 = 42.

4 Watch it reason

see the agent plan and solve, step by step.

The climax of the build: no single tool could solve it — the loop does. That loop is the agent.

Beyond a Single Agent

A short survey of where the field goes next — the single-agent skills you built transfer directly.

The Road Ahead — at a Glance

Evaluation

Judge the outcome (did it get done?) and the trajectory (were the steps sound?), plus reliability and safety.

Multi-agent

Several specialised agents collaborate, often coordinated by a supervisor agent.

Multimodal

Agents that can also see and hear — images, audio, video — not just read text.

Coding agents

Agents that run, test, and fix real code in a dedicated environment.

Surveyed briefly — not separate build modules.

Key Takeaways

1. Planning decomposes a goal into ordered sub-tasks.
2. ReAct loops THOUGHT -> ACTION -> OBSERVATION until a final answer.
3. A loop + a stopping rule (max_steps) is what makes an agent autonomous.
4. Reflection lets an agent critique and improve its own work.
5. Reasoning + Memory + Tools + Planning = a complete single agent.

Discussion

Q1 Name the three parts of a ReAct cycle and what each does.

Q2 What two things make an agent autonomous — and what does max_steps protect against?

Q3 Name the four components that together make a complete single agent.

That completes the agentic half — congratulations on building a full agent from scratch!